

01204211 Discrete Mathematics

Lecture 8a: Integers and GCD

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November 13, 2025

$\boxed{a}$ $\boxed{b}$

$a > b$

$\boxed{x}$ $\boxed{y}$



Twice

$\boxed{x+y}$ $\boxed{0}$

Once

$\boxed{a}$

$\boxed{\quad}$

$$y - (a - x) = y + x - a$$

$0, a, b, \dots$



$\boxed{0}$ $\boxed{x+y}$

$\boxed{a + b}$

$\boxed{\alpha a + \beta \cdot b}$

α, β integers

$\boxed{\quad}$ $\boxed{b}$

$$x - (b - y) = x + y - b$$

$\boxed{12}$ $\boxed{9}$

$\boxed{\alpha \cdot 12 + \beta \cdot 9}$

Number theory: integers and divisibility

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We will cover:

- ▶ Basic concepts of divisibility, prime numbers, and congruence.
- ▶ How to quickly check if a number is prime.
- ▶ How to essentially perform “division” with integers, allowing us to work with important and useful objects like polynomials using only integers.

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- ▶ Basic concepts of divisibility, prime numbers, and congruence.
- ▶ How to quickly check if a number is prime.
- ▶ How to essentially perform “division” with integers, allowing us to work with important and useful objects like polynomials using only integers.
- ▶ Applications like cryptography (RSA), secret sharing, erasure codes and error correcting codes.

Definitions

$$a \mid b$$

$$\frac{b}{a}$$

Definition (divisibility)

We say that an integer a divides b or b is divisible by a if there exist an integer k such that

$$b = ak.$$

If it is the case, we also write $a \mid b$. We also say that a is a divisor (or a factor) of b .
On the other hand if a does not divide b , we write $a \nmid b$.

Examples

If $a|b$ and $a|c$, prove that $a|(b+c)$.

• ၎င်းအတွက် $a|b$ ဖြစ်သည်ကို k နှင့်
 $b = ak$

• $a|c$ ဖြစ်သည်ကို l နှင့်
 $c = al$

• ထိုသို့ဆိုလျှင် $b+c = ak + al = (k+l)a$

ထို့ကြောင့် k နှင့် l ဖြစ်သည့် $k+l$ ဖြစ်သည်ကို m ဟုဆိုပါက

ထိုသို့ဆိုလျှင် $m = k+l$ နှင့်

$ma = b+c$ ဖြစ်သည်ကို

ဆိုလျှင် $a|b+c$ ဖြစ်သည်။

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Goal: $a|b$ ဖြစ်သည်ကို m နှင့် $b+c = ma$

Examples

If $a|b$ and $a|c$, prove that $a|(b+c)$.

If $a|b$ and $b|c$, prove that $a|c$.

สมมติ $a|b$ และ $a|c$ ให้ $b = k_1 a$
ให้ $c = k_2 b$
ดังนั้น $c = k_2 b = k_2 \cdot k_1 a$
ให้ k_1 และ k_2 เป็นจำนวนเต็ม
ดังนั้น $c = a \cdot (k_1 \cdot k_2)$
ให้ $c = a \cdot l$ ดังนั้น $a|c$ Q.E.D.

Remainder



Definition (remainder)

The **remainder** of the division of b with a is an integer r when there exists an integer q such that

where $0 \leq r < a$.

$$b = qa + r,$$
$$-47 = [-6] \cdot 9 + \overbrace{[7]}^{0, 1, \dots, 8}$$

$$47 \bmod 9 = 2$$

$$-47 \bmod 9 = (7) \quad \text{and} \quad \text{not } (-4)$$

$$347 \bmod 23 = 2$$

$$-347 \bmod 23 = (21)$$

Remainder

Defintion (remainder)

The remainder of the division of b with a is an integer r when there exists an integer q such that

$$b = qa + r,$$

(Handwritten notes: a pink circle around 'q', a pink arrow pointing to 'a' with the text '1xb', and a pink '=' above the equation)

where $0 \leq r < a$.

We refer to q as the quotient of the division.

(Handwritten red note: ↪ example)

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Examples:

We use operator mod to denote an operation for finding the remainder of a division.

I.e., $b \bmod a$ is the remainder of dividing b with a .

Examples

Let $\underline{r}$ be the remainder of the division of $\underline{b}$ by $\underline{a}$. Assume that $\underline{c}|\underline{a}$ and $\underline{c}|\underline{b}$. Prove that $\underline{c}|\underline{r}$.

$$\text{but } \underline{r} = b \bmod a, \text{ and } \underline{c}|\underline{a}, \underline{c}|\underline{b} \\ \Rightarrow \underline{c}|\underline{r}.$$

Proof: $\underline{b}$ is divided by $\underline{a}$ with remainder $\underline{r}$ and quotient $\underline{q}$.

$$\underline{b} = \underline{q} \cdot \underline{a} + \underline{r}$$

hence

$$\underline{r} = \underline{b} - \underline{q} \cdot \underline{a}$$

Since $\underline{c}|\underline{a}$, let $\underline{c}|\underline{b}$ also, then $\underline{a} = \underline{k}_1 \underline{c}$ and $\underline{b} = \underline{k}_2 \underline{c}$.

$$\text{Then we get } \underline{r} = \underline{k}_2 \underline{c} - \underline{q} \cdot \underline{k}_1 \underline{c} = (\underline{k}_2 - \underline{q} \cdot \underline{k}_1) \underline{c}$$

More examples

For every integer a , $a - 1 \mid a^2 - 1$.

Primes

Definition (primes)

- ▶ An integer $p > 1$ is a **prime** if its divisors are only p , $-p$, 1 , and -1 .
- ▶ If an integer $n > 1$ is not a prime, it is called a **composite**.
- ▶ Note: 1 is not a prime and also not a composite.

Fundamental theorem of arithmetic

$$15 = 3 \cdot 5 = 5 \cdot 3$$

$$3^2 \cdot 5^7 = \cancel{3^4 \cdot 2 \cdot 5}$$

Unique factorization

Every integer greater than 1 can be represented uniquely as a product of prime numbers, up to the order of the factors.

Algorithm for testing primes

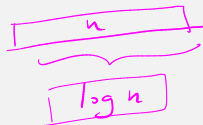
2, 3, 4, ..., n-2, n-1

Recall our CheckPrime2 algorithm

```
Algorithm CheckPrime2(n): // Input: an integer n
  if n <= 1:
    return False
  let s = square root of n
  i = 2
  while i <= s:
    if n is divisible by i:
      return False
    i = i + 1
  return True
```

$$n = p \cdot q$$

input n
input size



How fast can it run?

$O(\sqrt{n})$

$2^{\frac{\log n}{2}}$

bits

Algorithm for testing primes → cryptology.

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How fast can it run? Note that $s = \sqrt{n}$; therefore, it takes time $O(\sqrt{n})$ to run.

Efficient algorithms

Is $O(\sqrt{n})$ for checking a prime number efficient?

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When working with input consisting only a few numbers, we typically use the number of bits. For integer n , the number of bits of n is $\lceil \log_2 n \rceil$.

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n	number of bits of n	$\sqrt{n}$
2	1	1.414
4	2	2
16	4	4
1,024	10	32
1,048,576	20	1,024
1,125,899,906,842,624	50	33,554,432
1,267,650,600,228,229,401,496,703,205,376	100	1,125,899,906,842,624

Efficient algorithms

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Side note: Recall that the first step in RSA is to find a pair of large primes. Typically we want them to be of size in the *thousand* bits.

Greatest Common Divisors (GCD)

Definition (GCD)

≥ 1

For integers x and y , the **greatest common divisor** (or GCD) of x and y is the largest integer g such that $g|x$ and $g|y$. We refer to it as $\gcd(x, y)$.

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A simple way to find $\text{gcd}(x, y)$:

$$\boxed{\log x} + \boxed{\log y}$$

```
g = min(x,y)
while (x mod g != 0) or (y mod g != 0):
    g -= 1
return g
```

$$O(x)$$

$$x = 2^{\log x}$$

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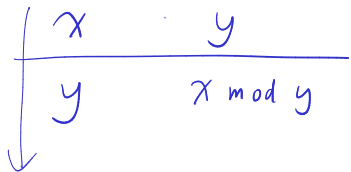
A simple way to find $\gcd(x, y)$:

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return g
```

What is the running time of this algorithm? Does it run in polynomial time on the size of the input?

Euclid's algorithm

```
Algorithm Euclid(x,y):  
  if  $x \bmod y == 0$ :  
    return y  
  else:  
    return Euclid(y,  $x \bmod y$ )
```



$$\begin{array}{r} 125, 100 \\ \hline 100 \quad (25) \end{array}$$

Euclid's algorithm

$O(\log^2 x)$

10.5.21.

✓ $\gcd(x, y) = \gcd(y, x \bmod y)$ when $x \bmod y \neq 0$.

Running time?

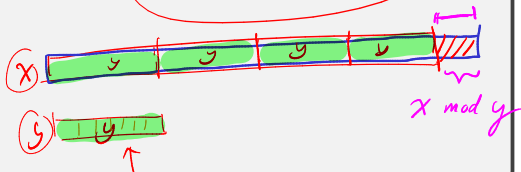
Algorithm Euclid(x, y):

if $x \bmod y == 0$:

return y ✓

else:

return Euclid(y, $x \bmod y$)



Let's see how it works with $\text{Euclid}(12311, 24324)$:

$\text{Euclid}(12311, 24324)$

$\text{Euclid}(24324, 12311)$

$\text{Euclid}(12311, 12013)$

$\text{Euclid}(12013, 298)$

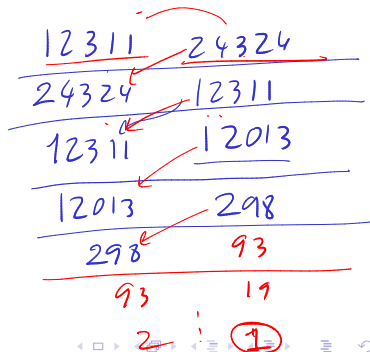
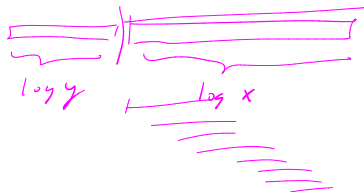
$\text{Euclid}(298, 93)$

$\text{Euclid}(93, 19)$

$\text{Euclid}(19, 17)$

$\text{Euclid}(17, 2)$

$\text{Euclid}(2, 1)$



We have to prove two properties:

- ▶ For any ^{positive} integers x and y , $\text{Euclid}(x, y) = \text{gcd}(x, y)$.
- ▶ The running time of Euclid.



Proofs

We have to prove two properties:

- ▶ For any integers x and y , $\text{Euclid}(x, y) = \text{gcd}(x, y)$.
- ▶ The running time of Euclid.

Note that when $x < y$, $\text{Euclid}(x, y)$ just calls itself with both arguments swapped, i.e., $\text{Euclid}(y, x)$. After that, in each call, x is always larger than y . For simplicity of the analysis, we shall work only with the case that $x > y$.

Theorem 1

For any integers x and y such that $x > y$, $\text{Euclid}(x, y) = \text{gcd}(x, y)$.

Proof.

We prove using strong induction. For the base case, note that when $y|x$, $\text{gcd}(x, y) = y$; ✓
therefore, the base case of the algorithm is correct.

Our induction hypothesis is: for any $x' < x$ and $y' < y$, $\text{Euclid}(x', y') = \text{gcd}(x', y')$.

Now assume that $y \nmid x$. The Euclid algorithm returns $\text{Euclid}(y, x \bmod y)$ as the gcd. Note that $y < x$ and $x \bmod y < y$. Therefore, we can use the I.H. to claim that

$$\rightarrow \text{Euclid}(y, x \bmod y) = \text{gcd}(y, x \bmod y).$$

Thus, we are left to show that

$$g = \text{gcd}(x, y)$$

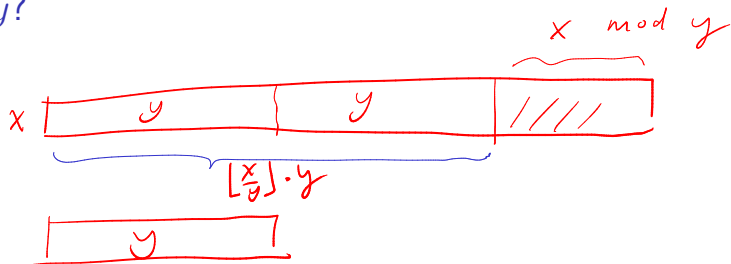
$$g' = \text{gcd}(y, x \bmod y)$$

$$\text{gcd}(x, y) = \text{gcd}(y, x \bmod y).$$

$$x - \left\lfloor \frac{x}{y} \right\rfloor \cdot y$$



What is $x \bmod y$?



$$x \bmod y = x - \left\lfloor \frac{x}{y} \right\rfloor \cdot y$$

What is $x \bmod y$?

Let $\lfloor a \rfloor$ be the largest integer a' such that $a' \leq a$.

floor,

$$\lfloor 0,5 \rfloor = 0$$

$$\lfloor -1,5 \rfloor = -2$$

$$\lfloor 10,7 \rfloor = 10$$

What is $x \bmod y$?

Let $\lfloor a \rfloor$ be the largest integer a' such that $a' \leq a$.

$$x \bmod y = x - \left\lfloor \frac{x}{y} \right\rfloor \cdot y$$

$$x \bmod y = x - \lfloor x/y \rfloor \cdot y$$

Lemma 2

If $a|x$ and $a|y$, then $a|x \bmod y$.

បើ $a|x$, $a|y$ ទាំង a រួច k_1, k_2 ទៅ

$$x = k_1 a, \quad y = k_2 a.$$

$$\text{ហើយ} \quad x \bmod y = x - \lfloor x/y \rfloor \cdot y$$

$$= k_1 a - \lfloor x/y \rfloor \cdot k_2 a = (k_1 - \lfloor x/y \rfloor \cdot k_2) \cdot a$$

បើ k_1, k_2, x, y ជាចំនួនគត់ ទាំង $k_1 - \lfloor x/y \rfloor \cdot k_2$ ក៏ជាចំនួនគត់ ដូច្នេះ a ចែក $x \bmod y$ ពិតជាបាន។ □

$$\text{បើ} \quad \gcd(x, y) | x$$

$$\gcd(x, y) | y$$

$$\Rightarrow \gcd(x, y) | x \bmod y$$

$$\triangleright g \leq g'$$

Lemma 2

If $a|x$ and $a|y$, then $a|x \bmod y$.

Lemma 3

✓ $\gcd(x, y) = \gcd(y, x \bmod y)$

Proof: ให้ $g = \gcd(x, y)$, ให้ $g' = \gcd(y, x \bmod y)$.

① จาก Lemma 2 , $g' \geq g$ เพราะว่า g เป็นตัวหารร่วมของ $y, x \bmod y$
bba'i g' เป็น ตัวหารร่วมที่มากที่สุด.

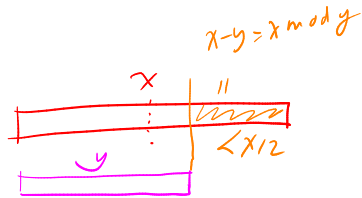
② จ: พิสูจน์ว่า $g' \leq g$ โดยแสดงว่า $g'|y, g'|x$

จะเห็นว่า

$$x = \underbrace{[x/y] \cdot y}_{\text{div}} + \underbrace{(x \bmod y)}_{\text{mod}}$$

bba'i $g'|y$ bba'i $g'|x \bmod y$
 $\Rightarrow g'|x$ ตาม

How many recursive calls does Euclid's algorithm makes?



$$\begin{aligned}x \bmod y \\&= x - y \\&< x/2\end{aligned}$$

Euclid(x, y)

Euclid($y, x \bmod y$)

Euclid($x \bmod y$, $y \bmod (x \bmod y)$)

Euclid($y \bmod (x \bmod y)$,
 $(x \bmod y) \bmod (y \bmod (x \bmod y))$)

if $y \leq x/2$ ✓

if $y > x/2$

How many recursive calls does Euclid's algorithm makes?

Consider Euclid(x, y):

$$y, x \bmod y$$

$\parallel$
 x

- If we start with $x < y$, the next calls will always have that $x > y$; so we have at most one call with $x < y$. ✓

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Consider $\text{Euclid}(x, y)$:

- ▶ If we start with $x < y$, the next calls will always have that $x > y$; so we have at most one call with $x < y$.
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- ▶ When can we decrease the value of x or y in the calls?
- ▶ When $y \leq x/2$, when $\text{Euclid}(x, y)$ calls $\text{Euclid}(y, x \bmod y)$ the first argument decreases by half.

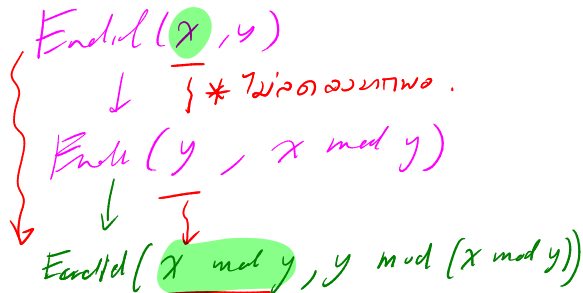
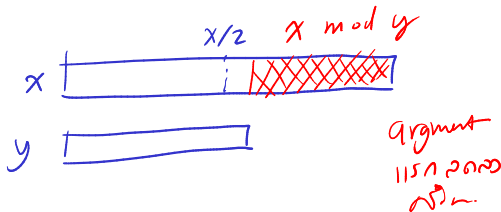
$\text{Euclid}(\underline{y}, y)$

$\text{Euclid}(\underline{y}, x \bmod y)$

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$\text{Euclid}(x, y) \Rightarrow \text{Euclid}(y, x \bmod y) \Rightarrow \text{Euclid}(x \bmod y, y \bmod (x \bmod y))$

$$\underline{x \bmod y} < \underline{x/2} \quad \text{if } y > x/2.$$

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 $\text{Euclid}(x, y) \Rightarrow \text{Euclid}(y, x \bmod y) \Rightarrow \text{Euclid}(x \bmod y, y \bmod (x \bmod y))$ Note that in this case, $x \bmod y = x - y \leq x/2$.

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Consider $\text{Euclid}(x, y)$:

- ▶ If we start with $x < y$, the next calls will always have that $x > y$; so we have at most one call with $x < y$.
- ▶ When can we decrease the value of x or y in the calls?
- ▶ When $y \leq x/2$, when $\text{Euclid}(x, y)$ calls $\text{Euclid}(y, x \bmod y)$ the first argument decreases by half.
- ▶ How about when $y > x/2$?

$\text{Euclid}(x, y) \Rightarrow \text{Euclid}(y, x \bmod y) \Rightarrow \text{Euclid}(x \bmod y, y \bmod (x \bmod y))$ Note that in this case, $x \bmod y = x - y \leq x/2$. Thus, after two recursive calls, the first argument decreases by half.

- ▶ How many times can that happen?
- ▶ The first argument can decrease by a factor of two for at most $\log x$ times.
Therefore, the Euclid algorithm runs in time $O(\log \max\{x, y\}) = O(\log x + \log y)$.

assum $x \geq y$

$$O(\log^3 x)$$

Computing power

$$x^2 \quad x^4 \quad (x^2)^2$$

$$x^{256} = ((((((x^2)^2)^2)^2)^2)^2)^2)^2$$

How fast can we compute x^y ?

```
[ out ← x  
  i ← 0  
  While i < k:  
    out ← (out * out) mod p.
```

$$x^{(2^k)}$$

```
def power(x, e, p):  
  if e == 1:  
    return x  
  else:  
    v ← power(x, e/2, p)  
    return (v * v) mod p
```

Computing power

How fast can we compute x^y ?

```
Algorithm power(x,y):  
  a = 1  
  for i = 1,2,...,y:  
    a *= x  
  return a
```

Computing power

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What is the running time?

Computing power

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Algorithm power(x,y):  
  a = 1  
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    a *= x  
  return a
```

What is the running time? Is it efficient?

Repeated squaring

If y is a power of two, we can find x^y using small number of multiplications using repeated squaring. E.g.,

$$x^{16} = (x^8)^2 = ((x^4)^2)^2 = (((x^2)^2)^2)^2.$$

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```
Algorithm power(x,y): // for  $y=2^k$   
  if  $y == 0$ :  
    return 1  
  else:  
     $a = \text{power}(x, y / 2)$   
    return  $a*a$ 
```

Repeated squaring (general y)

$$a = x^{\lfloor y/2 \rfloor}$$

Algorithm power(x,y):

if $y == 0$:

return 1

else:

a = power(x, floor(y / 2))

if $y \bmod 2 == 0$:

return a*a

else

return a*a*x

← odd y is q

$$a^2 = (x^{\lfloor y/2 \rfloor})^2 = x^y$$

$$a^2 = (x^{\lfloor y/2 \rfloor})^2 = (x^{y-1}) * x$$

Repeated squaring (general y)

```
Algorithm power(x,y):  
  if y == 0:  
    return 1  
  else:  
    a = power(x, floor(y / 2))  
    if y mod 2 == 0:  
      return a*a  
    else  
      return a*a*x
```

What is the number of recursive calls?

Repeated squaring (general y)

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Algorithm power(x,y):  
  if y == 0:  
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What is the number of recursive calls?

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      return a*a*x
```

What is the number of recursive calls?

What is the running time?

While the number of multiplication is small, the numbers involved is huge as x^y has $y \log x$ bits. Computing x^y exactly definitely takes a long time.

Repeated squaring (general y , mod n)

for return n , time $\underbrace{\log n}_{\text{bit}}$

Computing $x^y \bmod n$:

```
Algorithm power(x,y,n):  
  if y == 0:  
    return 1  
  else:  
    a = power(x, floor(y / 2)) mod n  
    if y mod 2 == 0:  
      return a*a mod n  
    else  
      return a*a*x mod n
```

$O((\log^2 n) \log y)$